Project Brief: UrbanFix (Group C)

1. Problem Statement & Target User Group

- **Target User Group:** Active urban citizens and daily commuters (drivers, cyclists, and pedestrians) who navigate city infrastructure regularly and care about safety and public maintenance.
- **The Task:** Quickly and effortlessly reporting infrastructural defects (such as potholes, malfunctioning streetlights, and damaged walkways) while on the move.
- **The Obstacle:** Current reporting channels rely on slow, fragmented bureaucracy (obscure municipal web forms, phone calls) that lack real-time geographical visibility and resolution tracking. This creates a discouraging "black hole" effect for citizens.

2. Competitive Analysis & Evidence-Based Differentiation

We evaluated existing platforms across Poland and internationally:

- **Italian/International Apps (e.g., WeDU!, IOparteciPA):** Many are strictly geofenced to specific municipalities, suffer from poor UX/UI leading to low app store reviews, or lack active maintenance.
- **miasto.app & NaprawmyTo.pl (Poland):** Web-based forms that simply forward reports to city offices via email. They lack dedicated native mobile performance, real-time public map visibility, and transparent status tracking.
- **mObywatel (Poland):** Official public app, but its reporting scope is strictly restricted to environmental violations (e.g., illegal dumping), omitting daily road hazards.
- **Local Municipal Portals (e.g., lckm.uml.lodz.pl/zgl in Łódź):** Emergency web pages that are cumbersome on mobile devices and closed to public viewing.
- **UrbanFix Differentiation:** UrbanFix provides a mobile-first, transparent public map that informs the community immediately, reduces duplicate reports, and tracks resolution status in real time.

3. Challenging Assumptions (AI & Peer Critique)

- **Assumption 1 (Data Quality):** *Users will submit accurate, constructive reports.*

Critique & Mitigation: Risk of spam or duplicate reports. For the MVP scope, validation relies on required GPS tagging and camera input; duplicate pin clustering can visually validate issue severity.

- **Assumption 2 (Privacy & Permissions):** *Users will willingly share camera and location access.*

Critique & Mitigation: Permission prompts must be transparent and contextual, requesting location and camera access only during active report creation.

- **Assumption 3 (Institutional Feedback):** *Reports will trigger prompt municipal actions.*

Critique & Mitigation: Without direct municipal API integration during development, ticket lifecycle states ("Submitted", "In Review", "Resolved") will be managed via a simulated local database.

4. Minimum Viable Product (MVP) Scope

Primary User Journey

Open application $\rightarrow$ Explore the local map and inspect active hazards $\rightarrow$ Tap "New Report" to capture a photo, add a brief description, and auto-detect GPS coordinates $\rightarrow$ Submit the report $\rightarrow$ Track ticket status inside the personal overview screen.

Core Screens & Features (3 Core Screens)

1. **Interactive Map / Feed Screen:** Displays active issue pins with category markers.
2. **Report Creation Screen:** Camera capture, automatic location pinning, description field, and a simplified category selector restricted to 4 macro-areas to streamline the MVP:
    a. *Roads & Mobility* (potholes, sidewalks, blockages)
    b. *Environment & Waste* (trash, vandalism)
    c. *Lighting & Safety* (broken streetlights, traffic signals)
    d. Other
3. **Ticket Detail / My Reports Screen:** Personal list of submitted tickets showing lifecycle badges (`Submitted`, `In Progress`, `Resolved`).

Explicitly Out of Scope

- Public user-to-user commenting sections and social forums.
- Gamification mechanics, user leaderboards, and achievement badges.
- Real-time municipal dispatch integration and worker route assignment.

5. Three User Stories & Acceptance Criteria

- **User Story 1 — Reporting Infrastructure Hazards**

As an active commuter, I want to submit a localized report with a photo and GPS tag, so that responsible teams know the exact location and nature of the hazard.

    - **Acceptance Criteria:** The application captures camera input and current coordinates, validates that required fields are not empty, and stores the ticket upon tapping "Submit".
- **User Story 2 — Consulting the Public Hazard Map**

As a city traveler, I want to view active issues on an interactive local map, so that I can anticipate road hazards and stay informed about my neighbourhood's condition.

    - **Acceptance Criteria:** The main screen displays interactive map markers corresponding to open tickets; tapping a marker presents a summary card with the photo and issue type.
- **User Story 3 — Tracking Report Status**

As a reporting citizen, I want to inspect the progress status of my submitted issues, so that I have clear transparency on whether the problem is being addressed.

    - **Acceptance Criteria:** The "My Reports" view presents each report alongside a visible status indicator (`Submitted`, `In Progress`, or `Resolved`).

6. Team Roles & Initial Task Allocation

- **UI/UX & Frontend Lead: Antonio** – Design and layout of Jetpack Compose screens (Map Screen, Report Form, Status Views).
- **Application Logic & Architecture Lead: Diego** – Data models, local state management, camera hardware integration, and GPS/location services.
- **Project Manager & Repository Lead: Efe** – Git repository hygiene, merging branches, conflict resolution, QA testing on emulator/devices, and submission documentation.